1 Introduction

The Kazhdan-Lusztig polynomial $P_M(t)$ is a fundamental invariant associated with any matroid M, as defined by Elias, Proudfoot, and Wakefield in [3]. This polynomial, denoted $P_M(t)$, exhibits formal similarities to the Kazhdan-Lusztig polynomials defined for Coxeter groups. The coefficients of $P_M(t)$ depend only on the lattice of flats $\mathcal{L}(M)$ of the matroid, and in fact, they are integral linear combinations of the flag Whitney numbers counting chains of flats with specified ranks.

In [2], Braden and Vysogorets presented a formula that relates the Kazhdan-Lusztig polynomial of a matroid M to that of the matroid obtained by deleting an element e, denoted $M \setminus e$, as well as various contractions and localizations of M. Specifically, for a simple matroid M where e is not a coloop, their main result, Theorem 2.8, states:

$$P_M(t) = P_{M\setminus e}(t) - tP_{M_e}(t) + \sum_{F\in S} \tau(M_{F\cup e}) \cdot t^{\mathrm{crk}(F)/2} \cdot P_{M^F}(t)$$

where the sum is taken over the set S of all subsets F of $E \setminus e$ such that both F and $F \cup e$ are flats of M, and $\tau(M)$ is the coefficient of $t^{(\mathrm{rk}(M)-1)/2}$ in $P_M(t)$ if $\mathrm{rk}(M)$ is odd, and zero otherwise.

The inverse Kazhdan-Lusztig polynomial $Q_M(x)$ is another important invariant. There is a related polynomial $\hat{Q}_M(x) = (-1)^{\mathrm{rk}(M)} Q_M(x)$ which acts as the inverse of the Kazhdan-Lusztig polynomial $P_M(t)$ within the incidence algebra of the lattice of flats $\mathcal{L}(M)$.

In this paper, we aim to prove the following deletion formula for $\hat{Q}_M(x)$:

Theorem 1 (Deletion Formula for $\hat{Q}_M(x)$)**.** *Let M be a simple matroid and e an element that is not a coloop. Then*

$$\hat{Q}_M(x) = \hat{Q}_{M\setminus e}(x) - (1+x)\cdot \hat{Q}_{M_e}(x) - \sum_{G\in S'} \tau(M_e^G) \cdot x^{\mathrm{rk}(G)/2} \cdot \hat{Q}_{M_G}(x)$$

where $S' = \{F \in \mathcal{L}(M) \mid e \in F \text{ and } F \setminus e \notin \mathcal{L}(M)\}$.

The proof of this theorem is the main goal of these notes.

2 Perverse elements and the KL basis

Let M be a matroid and $\mathcal{L}(M)$ be its lattice of flats.

- **The Module $\mathcal{H}(M)$**: Let $\mathcal{H} = \mathcal{H}(M)$ be the **free $\mathbb{Z}[t, t^{-1}]$-module with basis indexed by** $\mathcal{L}(M)$. Elements of $\mathcal{H}$ are formal sums of the form

$$\alpha = \sum_{F\in\mathcal{L}(M)} \alpha_F \cdot F, \quad \alpha_F \in \mathbb{Z}[t, t^{-1}].$$

- **The Abelian Subgroup $\mathcal{H}_p$**: $\mathcal{H}_p$ is an **abelian subgroup of** $\mathcal{H}$ consisting of all $\alpha \in \mathcal{H}$ such that for every flat $F \in \mathcal{L}(M)$, the following two conditions hold:

  i. $\alpha_F \in \mathbb{Z}[t]$.

ii. $\sum_{G \geq F} t^{\mathrm{rk}(F)-\mathrm{rk}(G)} \alpha_G \in Pal(0)$, where $Pal(0)$ is the set of Laurent polynomials $f(t)$ such that $f(t) = f(t^{-1})$.

- **The Elements ζ^F**: For any flat $F \in \mathcal{L}(M)$, an element $\zeta^F \in \mathcal{H}$ is defined as

$$\zeta^F = \sum_{G \leq F} t^{\mathrm{rk}(F)-\mathrm{rk}(G)} P_{M^F_G}(t^{-2}) \cdot G.$$

- **Basis of $\mathcal{H}_p$**: Proposition 2.13 of [2] states that the set of elements $\{\zeta^F\}_{F \in \mathcal{L}(M)}$ forms a **$\mathbb{Z}$-basis for** $\mathcal{H}_p$. Any element $\beta \in \mathcal{H}_p$ can be uniquely expressed as a linear combination of the ζ^F with integer coefficients:

$$\beta = \sum_{F \in \mathcal{L}(M)} \beta_F(0) \zeta^F.$$

This algebraic framework, involving the module $\mathcal{H}(M)$ and its subgroup $\mathcal{H}_p$ with the basis $\{\zeta^F\}$, provides a foundation for studying the Kazhdan–Lusztig polynomials of matroids, as demonstrated by its role in the derivation of deletion formulas.

3 The module homomorphism $\mathcal{H}(M) \to \mathcal{H}(M \setminus e)$

Let M be a simple matroid and e be an element of its ground set that is not a coloop. There is a surjective map $\mathcal{L}(M) \to \mathcal{L}(M \setminus e)$ sending a flat $F \in \mathcal{L}(M)$ to $F \setminus e \in \mathcal{L}(M \setminus e)$. There also exists a module homomorphism $\Delta : \mathcal{H}(M) \to \mathcal{H}(M \setminus e)$ which is $\mathbb{Z}[t, t^{-1}]$-linear and is defined on the basis elements $F \in \mathcal{L}(M)$ by

$$\Delta(F) = \begin{cases} F & \text{if } F \not\ni e \\ F \setminus e & \text{if } F \ni e \text{ and } F \setminus e \notin \mathcal{L}(M) \\ t^{-1} \cdot (F \setminus e) & \text{if } F \ni e \text{ and } F \setminus e \in \mathcal{L}(M). \end{cases}$$

This definition is taken from Section 2.6 of Braden and Vysogorets [2]. In [2], the authors also prove the following lemma:

Lemma 2. *Let $F \in \mathcal{L}(M)$ be a flat such that $e \in F$ and $F \setminus e \notin \mathcal{L}(M)$. Then*

$$\Delta(\zeta^F) = \zeta^{F \setminus e} + \sum_{G \in S(M^F)} \tau(M^F_{G \cup e}) \cdot \zeta^G$$

where $S(M^F) = \{G \in \mathcal{L}(M^F) \mid e \notin G \text{ and } G \cup e \in \mathcal{L}(M^F)\}$.

Theorem 3. *Let $F \in \mathcal{L}(M)$ be a flat such that $e \in F$ and $F \setminus e \in \mathcal{L}(M)$, then*

$$\Delta(\zeta^F) = (t + t^{-1}) \zeta^{F \setminus e}$$

where $\zeta^{F \setminus e}$ on the right-hand side is the ζ-element in $\mathcal{H}(M \setminus e)$ associated with the flat $F \setminus e \in \mathcal{L}(M \setminus e)$.

Proof. Let $F_0 = F \setminus e$. We are given $F_0 \in \mathcal{L}(M)$. By the definition of $\Delta(G)$:

- If $G \leq F$ and $e \notin G$: Then $G \leq F_0$. Since $G \in \mathcal{L}(M)$ and $e \notin G$, G is also a flat in $M \setminus e$, so $G \in \mathcal{L}(M \setminus e)$. In this case, $\Delta(G) = G$.
- If $G \leq F$ and $e \in G$: Let $G_0 = G \setminus e$. Then $G_0 \leq F_0$. Since $F_0 \in \mathcal{L}(M)$, it follows that $G_0 = G \cap F_0 \in \mathcal{L}(M)$. Thus $G_0 \in \mathcal{L}(M \setminus e)$. In this case, $\Delta(G) = t^{-1} G_0$.

Applying this to $\Delta(\zeta^F)$:

$$\begin{aligned}\Delta(\zeta^F) &= \Delta\left(\sum_{G\le F} t^{\mathrm{rk}_M(F)-\mathrm{rk}_M(G)} P_{M_G^F}(t^{-2})\cdot G\right)\\ &= \sum_{G\le F, G\neq e} t^{\mathrm{rk}_M(F)-\mathrm{rk}_M(G)} P_{M_G^F}(t^{-2})\cdot \Delta(G)\\ &\quad + \sum_{G\le F, G\ni e} t^{\mathrm{rk}_M(F)-\mathrm{rk}_M(G)} P_{M_G^F}(t^{-2})\cdot \Delta(G)\\ &= \sum_{G\le F_0} t^{\mathrm{rk}_M(F)-\mathrm{rk}_M(G)} P_{M_G^F}(t^{-2})\cdot G\\ &\quad + \sum_{\substack{G_0\le F_0\\ (G=G_0\cup\{e\})}} t^{\mathrm{rk}_M(F)-\mathrm{rk}_M(G_0\cup\{e\})} P_{M^F_{G_0\cup\{e\}}}(t^{-2})\cdot (t^{-1}G_0).\end{aligned}$$

We use the rank relations:

- $\mathrm{rk}_M(F) = \mathrm{rk}_{M\setminus e}(F_0) + 1$.
- For $G \le F_0$: $\mathrm{rk}_M(G) = \mathrm{rk}_{M\setminus e}(G)$.
- For $G_0 \le F_0$: $\mathrm{rk}_M(G_0\cup\{e\}) = \mathrm{rk}_{M\setminus e}(G_0) + 1$.

And the Kazhdan-Lusztig polynomial identities under these conditions:

- For $G \le F_0$: $M_G^F \cong (M\setminus e)_G^{F_0} \oplus U_{1,1}(\{e\})$, so $P_{M_G^F}(t^{-2}) = P_{(M\setminus e)_G^{F_0}}(t^{-2})$.
- For $G_0 \le F_0$: $M^F_{G_0\cup\{e\}} \cong (M\setminus e)^{F_0}_{G_0}$, so $P_{M^F_{G_0\cup\{e\}}}(t^{-2}) = P_{(M\setminus e)^{F_0}_{G_0}}(t^{-2})$.

Substituting these into the sums:

$$\begin{aligned}\Delta(\zeta^F) &= \sum_{G\le F_0} t^{(\mathrm{rk}_{M\setminus e}(F_0)+1)-\mathrm{rk}_{M\setminus e}(G)} P_{(M\setminus e)_G^{F_0}}(t^{-2})\cdot G\\ &\quad + t^{-1}\sum_{G_0\le F_0} t^{(\mathrm{rk}_{M\setminus e}(F_0)+1)-(\mathrm{rk}_{M\setminus e}(G_0)+1)} P_{(M\setminus e)^{F_0}_{G_0}}(t^{-2})\cdot G_0\\ &= t\sum_{G\le F_0, G\in\mathcal{L}(M\setminus e)} t^{\mathrm{rk}_{M\setminus e}(F_0)-\mathrm{rk}_{M\setminus e}(G)} P_{(M\setminus e)_G^{F_0}}(t^{-2})\cdot G\\ &\quad + t^{-1}\sum_{G_0\le F_0, G_0\in\mathcal{L}(M\setminus e)} t^{\mathrm{rk}_{M\setminus e}(F_0)-\mathrm{rk}_{M\setminus e}(G_0)} P_{(M\setminus e)^{F_0}_{G_0}}(t^{-2})\cdot G_0.\end{aligned}$$

Using the definition of $\zeta^{F_0}_{M\setminus e}$:

$$\begin{aligned}\Delta(\zeta^F) &= t\cdot\zeta^{F_0}_{M\setminus e} + t^{-1}\cdot\zeta^{F_0}_{M\setminus e}\\ &= (t+t^{-1})\zeta^{F_0}_{M\setminus e}.\end{aligned}$$

□

4 The Deletion Formula for $\hat{Q}_M(x)$

Lemma 4. *The standard basis $\{F\}_{F\in\mathcal{L}(M)}$ of $\mathcal{H}(M)$ satisfies*

$$F = \sum_{G\le F} t^{\mathrm{rk}(F)-\mathrm{rk}(G)} \hat{Q}_{M_G^F}(t^{-2})\cdot\zeta^G. \tag{1}$$

Proof of 1. Applying the homomorphism Δ to equation 1 when $F = E$, we get

$$E \setminus e = \sum_{G \leq E} t^{\mathrm{rk}(E)-\mathrm{rk}(G)} \hat{Q}_{M_G^E}(t^{-2}) \cdot \Delta(\zeta^G).$$

Using lemma 4, the coefficient of $\zeta^\emptyset$ in the left-hand side is $t^{\mathrm{rk}(E\setminus e)-\mathrm{rk}(\emptyset)} \cdot \hat{Q}_{M_\emptyset^{E\setminus e}}(t^{-2}) = t^{\mathrm{rk}(M)} \cdot \hat{Q}_{M\setminus e}(t^{-2})$.

By Theorem 3, the coefficient $\zeta^\emptyset$ on the right-hand side is non-zero only when $G \in \{\emptyset, e\} \cup S'$. In each of these cases, we have the following:

- For $G = \emptyset$, we have $\Delta(\zeta^\emptyset) = \zeta^\emptyset$.
- For $G = e$, we have $\Delta(\zeta^e) = (t + t^{-1})\zeta^\emptyset$.
- For $G \in S'$, we have $\Delta(\zeta^G) = \tau(M_e^G) \cdot \zeta^\emptyset +$ terms not including $\zeta^\emptyset$.

Collecting the coefficients of $\zeta^\emptyset$ on the right-hand side, we have:

$$\begin{aligned} &t^{\mathrm{rk}(E)-\mathrm{rk}(\emptyset)} \hat{Q}_{M_\emptyset^E}(t^{-2}) + t^{\mathrm{rk}(E)-\mathrm{rk}(e)}(t + t^{-1}) \hat{Q}_{M_e^E}(t^{-2}) + \sum_{G \in S'} t^{\mathrm{rk}(E)-\mathrm{rk}(G)} \hat{Q}_{M_G^E}(t^{-2}) \cdot \tau(M_e^G) \\ = &\, t^{\mathrm{rk}(M)} \hat{Q}_M(t^{-2}) + t^{\mathrm{rk}(M)-1}(t + t^{-1}) \hat{Q}_{M_e}(t^{-2}) + \sum_{G \in S'} t^{\mathrm{rk}(M)-\mathrm{rk}(G)} \hat{Q}_{M_G}(t^{-2}) \cdot \tau(M_e^G). \end{aligned}$$

Equating the coefficients of $\zeta^\emptyset$ on both sides, we obtain:

$$\hat{Q}_{M\setminus e}(t^{-2}) = \hat{Q}_M(t^{-2}) + (1 + t^{-2}) \hat{Q}_{M_e}(t^{-2}) + \sum_{G \in S'} t^{-\mathrm{rk}(G)} \hat{Q}_{M_G}(t^{-2}) \cdot \tau(M_e^G).$$

Finally, taking $x = t^{-2}$ and rearranging yields the desired statement:

$$\hat{Q}_M(x) = \hat{Q}_{M\setminus e}(x) - (1 + x) \cdot \hat{Q}_{M_e}(x) - \sum_{G \in S'} \tau(M_e^G) \cdot x^{\mathrm{rk}(G)/2} \cdot \hat{Q}_{M_G}(x).$$

□

The inverse Kazhdan-Lusztig polynomial $Q_M(x) = (-1)^{\mathrm{rk}(M)} \hat{Q}_M(x)$ then satisfies the following:

Corollary 5. *Let M be a simple matroid and e an element that is not a coloop. Then*

$$Q_M(x) = Q_{M\setminus e}(x) + (1 + x) \cdot Q_{M_e}(x) - \sum_{G \in S'} \tau(M_e^G) \cdot x^{\mathrm{rk}(G)/2} \cdot Q_{M_G}(x)$$

where $S' = \{F \in \mathcal{L}(M) \mid e \in F \text{ and } F \setminus e \notin \mathcal{L}(M)\}$.

Example 6. *Let* $\mathrm{M} := U_{m,d}$ *be the uniform matroid of rank d on $m + d$ elements with $d \geq 1$. Then*

$$Q_{U_{m,d}}(x) = Q_{U_{m-1,d}}(x) + (1 + x) \cdot Q_{U_{m,d-1}}(x) - \tau(U_{m,d-1}) \cdot x^{d/2}.$$

Proof. As none of the elements in the ground set are coloops, we may apply the deletion formula for a generic element e in the ground set. The set $\mathcal{T}_e$ then consists only of the top flat E and the sum over $\mathcal{T}_e$ reduces to

$$\tau(M_e) \cdot x^{\mathrm{rk_M}(M)/2} = \tau(U_{m,d-1}) \cdot x^{d/2}.$$

□

Example 7. *Let* M *be the parallel connection of the cycle matroids $U_{1,m}$ and $U_{1,n}$ along the element i. Then*

$$Q_{\mathrm{M}}(x) = Q_{U_{1,m+n-1}}(x) + (1+x)\cdot Q_{U_{1,m-1}}(x)\cdot Q_{U_{1,n-1}}(x) - \left[\tau(U_{1,m-1})\cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1})\cdot x^{n/2} Q_{U_{1,m-1}}(x)\right].$$

Specifically, when the ranks m and n are odd, the τ terms are 0 and hence the formula simplifies to:

$$Q_{\mathrm{M}}(x) = Q_{U_{1,m+n-1}}(x) + (1+x)\cdot Q_{U_{1,m-1}}(x)\cdot Q_{U_{1,n-1}}(x).$$

Proof. We first note that for the parallel connection M of $U_{1,m}$ and $U_{1,n}$ along the element i, the set $\mathcal{T}_i$ consists of the top flat E and the two flats $F \cong U_{1,m}$ and $G \cong U_{1,n}$. Thus the sum over $\mathcal{T}_i$ reduces to:

$$\begin{aligned}&\tau(\mathrm{M}_i^E)\cdot x^{\mathrm{rk}(E)/2} Q_{\mathrm{M}_E}(x) + \tau(\mathrm{M}_i^F)\cdot x^{\mathrm{rk}(F)/2} Q_{\mathrm{M}_F}(x) + \tau(\mathrm{M}_i^G)\cdot x^{\mathrm{rk}(G)/2} Q_{\mathrm{M}_G}(x)\\ =&\tau(\mathrm{M}_i^E)\cdot x^{(m+n-1)/2} + \tau(U_{1,m-1})\cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1})\cdot x^{n/2} Q_{U_{1,m-1}}(x).\end{aligned}$$

Since $\mathrm{M}_i^E \cong U_{1,m-1}\oplus U_{1,n-1}$, we have $\tau(\mathrm{M}_i^E) = 0$ by Lemma 2.7 in [1]. Using $Q_{\mathrm{M}\setminus i}(x) = Q_{U_{1,m+n-1}}(x)$ and $Q_{\mathrm{M}_i}(x) = Q_{U_{1,m-1}\oplus U_{1,n-1}}(x) = Q_{U_{1,m-1}}(x)\cdot Q_{U_{1,n-1}}(x)$, we obtain:

$$Q_{\mathrm{M}}(x) = Q_{U_{1,m+n-1}}(x) + (1+x)\cdot Q_{U_{1,m-1}}(x)\cdot Q_{U_{1,n-1}}(x) - \left[\tau(U_{1,m-1})\cdot x^{m/2} Q_{U_{1,n-1}}(x) + \tau(U_{1,n-1})\cdot x^{n/2} Q_{U_{1,m-1}}(x)\right].$$

□

Example 8. *Let $M = PG(r-1,q)$ be the projective geometry of rank r over the finite field $GF(q)$, for $r \geq 2$. Then the deletion formula for its inverse Kazhdan-Lusztig polynomial is:*

$$Q_M(x) = Q_{M\setminus e}(x) + (1+x)\cdot Q_{PG(r-2,q)}(x) - \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x)$$

where $\begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q = \frac{q^{r-1}-1}{q-1}$ is the number of lines through the point e in M.

Proof. We start from the formula for $Q_M(x)$ in Theorem **??** and substitute $M = PG(r-1,q)$. The lattice of flats of a projective geometry is modular. For any matroid N with a modular lattice of flats, its Kazhdan-Lusztig polynomial $P_N(t)$ is 1. [4][Proposition 2.14] The coefficient $\tau(N)$ is the coefficient of $t^{(\mathrm{rk}(N)-1)/2}$ in $P_N(t)$, which is zero if $\mathrm{rk}(N) \geq 2$.

The sum in Theorem **??** is over the set $S' = \{G \in \mathcal{L}(M) \mid e \in G \text{ and } G \setminus e \notin \mathcal{L}(M)\}$. For the projective geometry M, a flat G containing e is a linear subspace. If $\mathrm{rk}(G) \geq 2$, removing the point e results in a set $G \setminus e$ that is not a subspace, and thus not a flat. If $\mathrm{rk}(G) = 1$, then $G = \{e\}$ and $G \setminus e = \emptyset$, which is a flat. Thus, for $M = PG(r-1,q)$, the set S' consists of all flats containing e of rank 2 or greater.

For each $G \in S'$, the term in the sum is $\tau(M_e^G)\cdot x^{\mathrm{rk}(G)/2}\cdot Q_{M_G}(x)$. The matroid M_e^G is the interval $[e,G]$ isomorphic to $PG(k-2,q)$ where $k = \mathrm{rk}(G)$. The rank of M_e^G is $k-1$ and $\tau(M_e^G)$, therefore, is non-zero only if the rank $k-1 = 1$, which implies $k = 2$.

Thus, the sum is over the set of rank-2 flats containing the point e. The number of such flats is the number of 2-dimensional subspaces of $V(r,q)$ containing a given 1-dimensional subspace, which is equal to the number of 1-dimensional subspaces in the quotient space $V(r,q)/\langle e\rangle \cong V(r-1,q)$. This number is $\begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q$. For each such rank-2 flat G:

- $\tau(M_e^G) = \tau(PG(0,q)) = 1$ since $PG(0,q)$ has rank 1.
- $Q_{M_G}(x) = Q_{PG(r-3,q)}(x)$, since the contraction M_G of $M = PG(r-1,q)$ by a rank-2 flat G is isomorphic to $PG(r-3,q)$.

The entire sum therefore reduces to:

$$\sum_{G \in S', \mathrm{rk}(G)=2} \tau(M_e^G) \cdot x^{\mathrm{rk}(G)/2} \cdot Q_{M_G}(x) = \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x).$$

Substituting this and the fact that the contraction M_e is isomorphic to $PG(r-2,q)$ back into the general formula from Corollary 5, we get:

$$Q_{PG(r-1,q)}(x) = Q_{M\setminus e}(x) + (1+x) \cdot Q_{PG(r-2,q)}(x) - \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q \cdot x \cdot Q_{PG(r-3,q)}(x). \quad \square$$

$\square$

Remark. The inverse Kazhdan-Lusztig polynomial of a projective geometry $PG(k-1,q)$ of rank k is known to be the constant polynomial $q^{\binom{k}{2}}$. This allows for substituting the known values for the Q polynomials in the formula to obtain an explicit expression for $Q_{M\setminus e}(x)$. Rearranging the terms and substituting gives:

$$\begin{aligned} Q_{M\setminus e}(x) &= Q_{PG(r-1,q)}(x) - (1+x)Q_{PG(r-2,q)}(x) + \begin{bmatrix} r-1 \\ 1 \end{bmatrix}_q x\, Q_{PG(r-3,q)}(x) \\ &= q^{\binom{r}{2}} - (1+x)q^{\binom{r-1}{2}} + \frac{q^{r-1}-1}{q-1} q^{\binom{r-2}{2}} \cdot x. \end{aligned}$$

5 The Deletion Formula for $\hat{Y}_M(x)$

The polynomial $\hat{Y}_M(x)$ is given by:

$$\hat{Y}_M(x) = \sum_{F \in \mathcal{L}(M)} \left(x^{\mathrm{rk}(F)} \mu[\emptyset, F] \right) \cdot \hat{Q}_{M_F}(x) = \sum_{F \in \mathcal{L}(M)} \left(x^{\mathrm{crk}(F)} \mu[F, E] \right) \cdot \hat{Q}_{M^F}(x).$$

5.1 suggestion 1

Define $\mu_F = \sum_{G \leq F} x^{\mathrm{rk}(G)} \mu[G,F] \cdot G$. Thus

$$\mu_M = \sum_{F \in \mathcal{L}(M)} \left(x^{\mathrm{rk}(F)} \mu[F, E] \right) \cdot \left(\sum_{G \leq F} x^{\mathrm{rk}(F) - \mathrm{rk}(G)} \hat{Q}_{M_G^F}(x^{-2}) \cdot \zeta^G \right)$$

5.2 take 2

Following [1] section 2.8, define a linear map $\psi : \mathcal{H}(M) \to \mathbb{Z}[x, x^{-1}]$ by

$$\zeta^F \mapsto x^{-\mathrm{rk}(F)} \mu[\emptyset, F].$$

Then, for $F \in \mathcal{L}(M)$,

$$\begin{aligned}\psi(F) &= \psi\left(\sum_{G\le F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\hat{Q}_{M^F_G}(x^{-2})\cdot\zeta^G\right)\\ &= \sum_{G\le F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\hat{Q}_{M^F_G}(x^{-2})\cdot x^{-\mathrm{rk}(G)}\mu[\emptyset,G]\\ &= x^{\mathrm{rk}(F)}\hat{Y}_{M^F}(x^{-2}).\end{aligned}$$

Unfortunately, the analogous relation $\psi_{M\setminus e}\circ\Delta = \psi_M$ does not hold.

Applying $\psi\circ\Delta$ to $E = \sum_{F\in\mathcal{L}(M)} x^{\mathrm{rk}(E)-\mathrm{rk}(F)}\hat{Q}_{M_F}(x^{-2})\cdot\zeta^F$, we get

$$\begin{aligned}x^{\mathrm{rk}(E\setminus e)}\hat{Y}_{M\setminus e}(x) &= \sum_{F\in\mathcal{L}(M)} x^{\mathrm{rk}(E)-\mathrm{rk}(F)}\hat{Q}_{M_F}(x^{-2})\cdot\psi\circ\Delta(\zeta^F)\\ \text{or,}\qquad \hat{Y}_{M\setminus e}(x) &= \sum_{F\in\mathcal{L}(M)} x^{-\mathrm{rk}(F)}\hat{Q}_{M_F}(x^{-2})\cdot\psi\circ\Delta(\zeta^F)\end{aligned}$$

We need to show that the right-hand side is equal to

$$\begin{aligned}&\hat{Y}_M(x^{-2}) - (1+x^{-2})\hat{Y}_{M_e}(x^{-2}) + \sum_{G\in S'}\tau(M^G_e)\cdot x^{-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2})\\ =&\sum_{F\in\mathcal{L}(M)} x^{-2\mathrm{rk}(F)}\mu[\emptyset,F]\cdot\hat{Q}_{M_F}(x^{-2}) - (1+x^{-2})\sum_{F\ge e} x^{-2\mathrm{rk}(F)+2}\mu[e,F]\cdot\hat{Q}_{M_F}(x^{-2})\\ &+\sum_{G\in S'}\tau(M^G_e)\cdot x^{-\mathrm{rk}(G)}\sum_{F\ge G} x^{-2\mathrm{rk}(F)+2\mathrm{rk}(G)}\mu[G,F]\cdot\hat{Q}_{M_G}(x^{-2})\end{aligned}$$

Case for ζ^F	$\Delta(\zeta^F)$	$\psi\circ\Delta(\zeta^F)$ **in terms of** $M\setminus e$	**Expression in terms of** M
$F \not\ni e$	ζ^F (in $\mathcal{H}(M\setminus e)$)	$t^{-\mathrm{rk}_{M\setminus e}(F)}\mu_{(M\setminus e)^F}$	$t^{-\mathrm{rk}_M(F)}\mu_{M^F}$
$F \ni e$ and $F\setminus e \in \mathcal{L}(M)$	$(t+t^{-1})\zeta^{F\setminus e}$	$(t+t^{-1})t^{-\mathrm{rk}_{M\setminus e}(F\setminus e)}\mu_{(M\setminus e)^{F\setminus e}}$	$(t^2+1)t^{-\mathrm{rk}_M(F)}\mu_{M^{F\setminus e}} =$ $-(t^2+1)t^{-\mathrm{rk}_M(F)}\mu_{M^F}$
$F \ni e$ and $F\setminus e \notin \mathcal{L}(M)$	$\zeta^{F\setminus e} + \sum_{H\in S(M^F)}\tau(M^F_{H\cup e})\zeta^H$	$t^{-\mathrm{rk}_{M\setminus e}(F\setminus e)}\mu_{(M\setminus e)^{F\setminus e}} +$ $\sum\tau(\dots)t^{-\mathrm{rk}H}\mu_{(M\setminus e)^H}$	$t^{-\mathrm{rk}_M(F)}\mu_{M^{F\setminus e}} +$ $\sum_{H\in S(M^F)}\tau(M^F_{H\cup e})t^{-\mathrm{rk}H}\mu_{M^H}$

Table 1: Action on the KL Basis

6 The Deletion Formula for $\hat{Y}_M(x)$ - proof

In order to prove the deletion for $Y_M(t)$, we consider the related polynomial $\hat{Y}_M(x)$ is given by:

$$\hat{Y}_M(x) = \sum_{F\in\mathcal{L}(M)}\left(x^{\mathrm{crk}(F)}\mu[F,E]\right)\cdot\hat{Q}_{M^F}(x).$$

For $G \in \mathcal{L}(M)$, we have

$$\hat{Y}_{M_G}(x) = \sum_{F \geq G} x^{\mathrm{crk}(F)} \mu[F, E] \cdot \hat{Q}_{M_G^F}(x).$$

We define $\mu^F = \sum_{G \leq F} x^{\mathrm{rk}(G)} \mu_M[G, F] \cdot G$ as our new basis for $\mathcal{H}(M)$ and consider its image under the deletion map Δ. Here, $\mu_M[G, F]$ is the Moebius function of the interval $[G, F]$ in the lattice of flats of M.

Lemma 9. *If e is not a coloop of M, then*

$$\Delta(\mu^E) = \mu^{E \setminus e}.$$

Proof. We have

$$\Delta(\mu^E) = \sum_{G \leq E} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G).$$

We partition the lattice of flats $\mathcal{L}(M)$ into four disjoint sets:

- $S = \{G \in \mathcal{L}(M) \mid e \notin G \text{ and } G \cup e \in \mathcal{L}(M)\}$.
- $T = \{G \in \mathcal{L}(M) \mid e \in G \text{ and } G \setminus e \in \mathcal{L}(M)\}$.
- $A = \{G \in \mathcal{L}(M) \mid e \notin G \text{ and } G \cup e \notin \mathcal{L}(M)\}$.
- $B = \{G \in \mathcal{L}(M) \mid e \in G \text{ and } G \setminus e \notin \mathcal{L}(M)\}$.

We can now rewrite the sum by splitting it over these four sets:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{G \in S} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) + \sum_{G \in T} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G) + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot \Delta(G).\end{aligned}$$

Applying the definition of Δ to each sum:

- For $G \in S \cup A$ (flats not containing e), $\Delta(G) = G$.
- For $G \in T$, $\Delta(G) = x^{-1}(G \setminus e)$.
- For $G \in B$, $\Delta(G) = G \setminus e$.

Substituting these into the expression gives:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{G \in S} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in T} x^{\mathrm{rk}(G)-1} \mu[G, E] \cdot (G \setminus e) \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot (G \setminus e).\end{aligned}$$

There is a natural bijection between S and T where for each $F \in S$, we have $F \cup e \in T$. We re-index the sum over T using $F \in S$, which lets us combine the first two sums:

$$\begin{aligned}\Delta(\mu_E) = &\sum_{F \in S} \left(x^{\mathrm{rk}(F)} \mu[F, E] + x^{\mathrm{rk}(F \cup e)-1} \mu[F \cup e, E] \right) \cdot F \\ &+ \sum_{G \in A} x^{\mathrm{rk}(G)} \mu[G, E] \cdot G + \sum_{G \in B} x^{\mathrm{rk}(G)} \mu[G, E] \cdot (G \setminus e).\end{aligned}$$

For $F \in S$, $\mathrm{rk}(F \cup e) = \mathrm{rk}(F) + 1$. Further, since e is not a loop or coloop of the minor M/F, we have the standard deletion-contraction relation for the Möbius function:

$$\mu_M[F, E] + \mu_M[F \cup e, E] = \mu_{M\setminus e}[F, E \setminus e].$$

We have analogous relations for the flats in the other sets. For a flat $G \in A$, the interval $[G, E]$ in $\mathcal{L}(M)$ is isomorphic to the interval $[G, E \setminus e]$ in $\mathcal{L}(M \setminus e)$, which gives:

$$\mu_M[G, E] = \mu_{M\setminus e}[G, E \setminus e].$$

For a flat $G \in B$, its corresponding set $F = G \setminus e$ is a flat in $M \setminus e$ but not in M. The intervals $[G, E]$ and $[G \setminus e, E \setminus e]$ are isomorphic, so the relationship between the Möbius functions in this case is:

$$\mu_M[G, E] = \mu_{M\setminus e}[G \setminus e, E \setminus e].$$

Substituting these identities into the expression for $\Delta(\mu^E)$ yields:

$$\begin{aligned}\Delta(\mu^E) = &\sum_{F\in S} x^{\mathrm{rk}(F)}\mu_{M\setminus e}[F, E \setminus e] \cdot F \\ &+ \sum_{G\in A} x^{\mathrm{rk}(G)}\mu_{M\setminus e}[G, E \setminus e] \cdot G + \sum_{G\in B} x^{\mathrm{rk}(G)}\mu_{M\setminus e}[G \setminus e, E \setminus e] \cdot (G \setminus e).\end{aligned}$$

Since the terms on the right account for all the flats in $\mathcal{L}(M \setminus e)$, we have:

$$\Delta(\mu^E) = \mu^{E\setminus e}.$$

□

Writing out μ^E in terms of Kazhdan-Lusztig basis elements, we have:

$$\begin{aligned}\mu^E &= \sum_{F\in\mathcal{L}(M)} x^{\mathrm{rk}(F)}\mu[F, E]\left(\sum_{G\le F} x^{\mathrm{rk}(F)-\mathrm{rk}(G)}\hat{Q}_{M_G^F}(x^{-2}) \cdot \zeta^G\right) \\ &= \sum_{F\in\mathcal{L}(M)}\sum_{G\le F} x^{2\mathrm{rk}(F)-\mathrm{rk}(G)}\mu[F, E] \cdot \hat{Q}_{M_G^F}(x^{-2}) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(M)}\sum_{F\ge G} x^{2\mathrm{rk}(F)-\mathrm{rk}(G)}\mu[F, E] \cdot \hat{Q}_{M_G^F}(x^{-2}) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(M)} x^{2\mathrm{rk}(E)-\mathrm{rk}(G)}\left(\sum_{F\ge G} x^{2\mathrm{rk}(F)-2\mathrm{rk}(E)}\mu[F, E]\hat{Q}_{M_G^F}(x^{-2})\right) \cdot \zeta^G \\ &= \sum_{G\in\mathcal{L}(M)} x^{2\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2}) \cdot \zeta^G\end{aligned}$$

Applying the deletion map Δ to this expression and using the lemma, we get:

$$\begin{aligned}\Delta(\mu^E) &= \sum_{G\in\mathcal{L}(M)} x^{2\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2}) \cdot \Delta(\zeta^G) \\ \text{or,}\qquad \mu^{E\setminus e} &= \sum_{G\in\mathcal{L}(M)} x^{2\mathrm{rk}(E\setminus e)-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2}) \cdot \Delta(\zeta^G)\end{aligned}$$

The coefficient of $\zeta^{\emptyset}$ on the left is $x^{2\mathrm{rk}(E\setminus e)}\hat{Y}_{M\setminus e}(x^{-2})$. As in the proof of Theorem 1, the coefficient of $\zeta^{\emptyset}$ on the right is:

$$x^{2\mathrm{rk}(E)}\hat{Y}_M(x^{-2}) - (x+x^{-1})x^{2\mathrm{rk}(E)-1}\hat{Y}_{M_e}(x^{-2}) + \sum_{G\in S'}\tau(M_e^G)\cdot x^{2\mathrm{rk}(E)-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2}).$$

Equating the coefficients of $\zeta^{\emptyset}$ on both sides, we get:

$$\hat{Y}_{M\setminus e}(x^{-2}) = \hat{Y}_M(x^{-2}) - (1+x^{-2})\hat{Y}_{M_e}(x^{-2}) + \sum_{G\in S'}\tau(M_e^G)\cdot x^{-\mathrm{rk}(G)}\hat{Y}_{M_G}(x^{-2}).$$

Substituting $t = x^{-2}$, we get:

$$\hat{Y}_{M\setminus e}(t) = \hat{Y}_M(t) - (1+t)\hat{Y}_{M_e}(t) + \sum_{G\in S'}\tau(M_e^G)\cdot x^{\mathrm{rk}(G)/2}\hat{Y}_{M_G}(t).$$

Using the fact that $\hat{Y}_M(t) = (-1)^{\mathrm{rk}(M)}Y_M(t)$, we get the following formula:

Theorem 10. *Let M be a matroid and e be a non-coloop of M. Then,*

$$Y_M(t) = Y_{M\setminus e}(t) - (1+t)Y_{M_e}(t) - \sum_{G\in S'}\tau(M_e^G)\cdot x^{\mathrm{rk}(G)/2}Y_{M_G}(t).$$

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